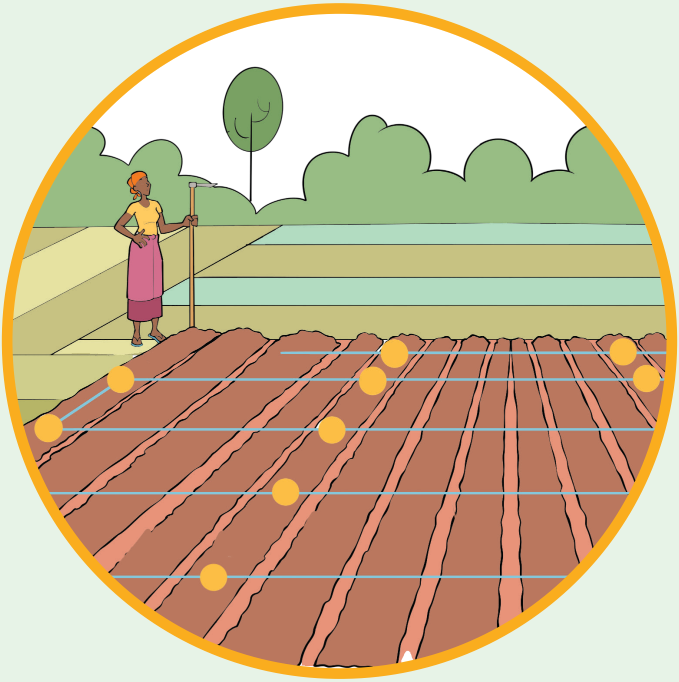
INTEGRATED TRAINING MANUALS ON SOIL, CROP AND PEST MANAGEMENT

GanzAfrica

SOIL SAMPLING TRAINING MANUAL



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Introduction

Careful soil sampling is critical for accurate soil analyses and reliable nutrient recommendations. If soil samples are not collected properly, one cannot expect the soil test results to provide accurate information about the nutrient or mineral status of the soil. The analysis can only be as good as the sample. Understanding the condition of your soil is a critical step before any planting activity. This manual is designed to guide you through the correct soil sampling procedures, ensuring that the test results accurately reflect the true condition of your field. Following the best practices will help maximise productivity and promote sustainable land use.

When to Sample

- It is recommended to collect soil samples on a sunny day.
- During pre-season for soil fertility assessment.
- It is recommended to test your soils every 4 years of production

Step 1 for Soil Testing: Define the purpose. Why Test the Soil?

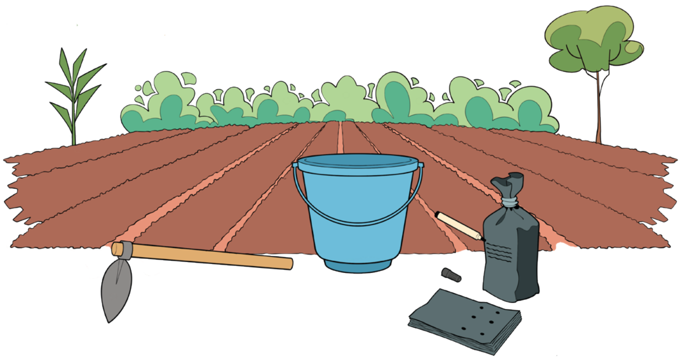
- Identify nutrient deficiencies
- Assess soil fertility
- Check soil pH (acidity or alkalinity)
- Choose the right fertilisers
- Improve land productivity and reduce input waste
- Informs the crop choice

- What does my soil need?
- How can I increase my yield?
- Which fertiliser is right for me?

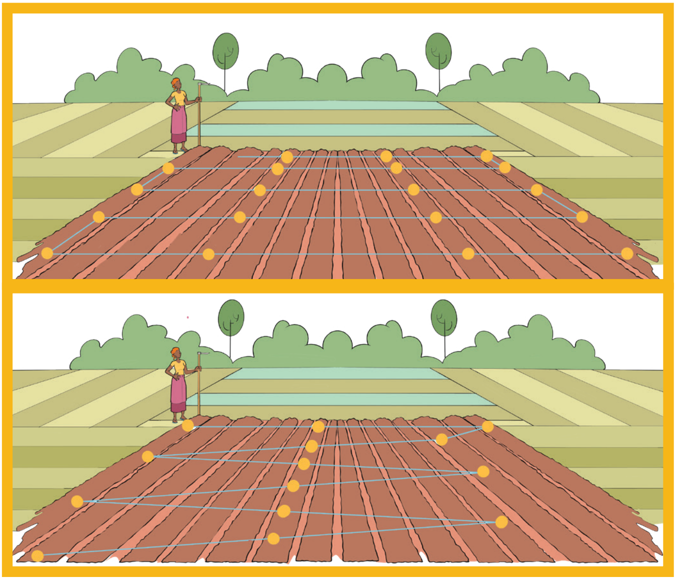
Step 2 for Soil Testing: Collecting Samples, Packaging, and Labelling

Materials Required:

A bucket, a shovel, plastic bags, and a permanent marker



- **To collect desirable samples,** walk in a zigzag pattern or grid pattern across the field to collect the samples. Always make sure that the collecting spots are distributed across the field.



- At each spot
  - Clear surface debris.
  - Using a spade/ shovel, dig a V-shaped 30 cm deep hole
  - Collect the soil vertically across the 30 cm profile and put the dug soil in a bucket.

- Mix the collected soil from this individual point in a bucket thoroughly to get a representative sample.
- From the mixed soil (bucket), take two handfuls/ one 500-gram cup of the mixed soil and place them into a plastic bag/ Khaki bag and tightly seal the bag.
- Go to the next spot and repeat the process across the field
- Once you have collected samples from all desired spots, label the bags with
  - Your names
  - Plot number/ID
  - Sample number/name/marker
  - Date
  - Put a marker with a name for each spot where you collected the soil sample. Alternatively, use GPS location (if available)
  - Observe for soil colour differences, water drainage, and make notes.

Step 3 for Soil Testing: Transport samples to the testing labs

After collecting soil samples, do not delay in submitting the samples to the lab for analysis. Below are the testing labs in Rwanda. Always coordinate with the sector agronomist to assist in submitting the samples.

Recommended testing laps in Rwanda

1.Rubona RAB Station – Huye

- Location: Huye
- Contact: +250 783 502 100
- The soil testing price starts from RWF 21 500 per soil sample

2.NAEB Laboratory – Kigali

- Location: Kigali
- Contact: +250 783 502 100
- The soil testing price starts from RWF 36,600 per soil sample

NB: Contact your sector/district agronomist for assistance on submitting your soil samples to a lab

Step 4 for Soil Testing: Soil Results Interpretation

The lab will send the analysis report after one to two weeks. It is recommended to seek the guidance of the sector agronomist to interpret the results. Below is a simple guide for results interpretation.

Number	What is Tested	What it Means
1	pH (soil acidity)	<6 = acidic; 6–6.5 = good; >7 = alkaline
2	Organic matter	5–8% or more = good soil
3	Nitrogen (N)	50–125 = good for most soils
4	Phosphorus	30–75 = good for crops
5	CEC (Nutrient Holding)	>12 = holds nutrients well; <8 = sandy, low holding

References:

- An agronomist will help you read your results and decide what to do next. Here's a simple guide: (Adapted from Michigan State University, 2016)
- National Agricultural Export Development Board https://www.naeb.gov.rw/
- Rwanda Agriculture and Animal Resources Development Board (RAB). https://www.rab.gov.rw/
- Rwanda Agriculture and Animal Resources Development Board. https://www.rab.gov.rw/
- Rwanda Soil Information Service (RwaSIS). https://rwasis.rab.gov.rw/
- D. Warncke,2016. Michigan State University. Understanding the MSU Soil Test Report. https://www.canr.msu.edu/resources/understanding_the_msu_soil_test_report_e0015#:~:text=As%20the%20soil%20test%20values,less%20than%20the%20potassium%20percentage

SOIL FERTILITY MANAGEMENT TRAINING MANUAL

Introduction

Soil fertility management is essential for improving good crop health and yield maximisation. Climate-smart agriculture enhances these outcomes through low-cost, sustainable practices that build soil health, improve soil moisture retention, and protect the environment. This manual promotes the use of CA practices for smallholder farmers using locally available resources.

Conservation agriculture (CA) is a climate-smart, resource-efficient way of farming that improves productivity while protecting soil and the environment. Unlike conventional farming methods that may degrade the land over time, conservation agriculture promotes soil health, water conservation, and resilience to climate shocks such as droughts. Planting climbing beans using minimum tillage and proper spacing boosts productivity and maintains soil health. This manual outlines the best practices for this method.

Why Conservation Agriculture?

- Improve crop yields over time by enhancing soil fertility
- Reduces labour and fuel costs by minimising ploughing
- Protects the soils from erosion, drying out, and nutrient loss
- Retains soil moisture and reduces water needs
- Encourages biodiversity and beneficial soil organisms

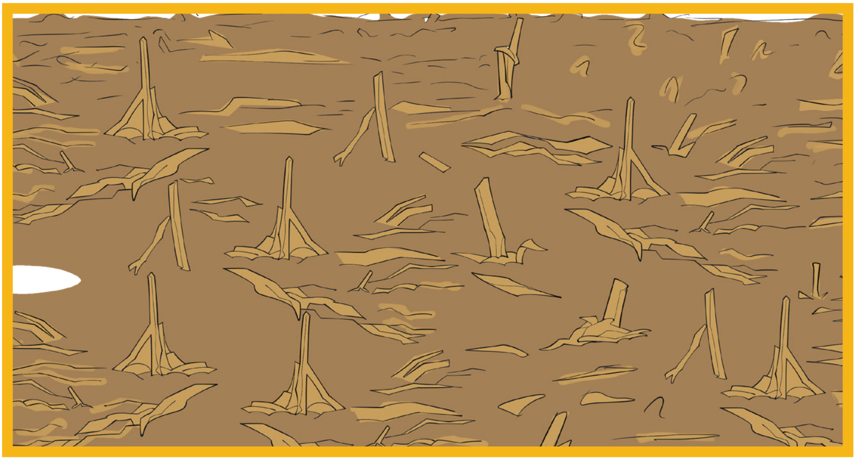
Principles of Conservation Agriculture

1. Minimum tillage

- Only dig planting holes/ basins where seeds will be planted. This helps the soil hold more water and reduces erosion.

2. Permanent soil cover

- Cover the soil with mulch (like dry grass or maize stalks) or cover crops (like mucuna). This keeps moisture in the soil and prevents erosion.



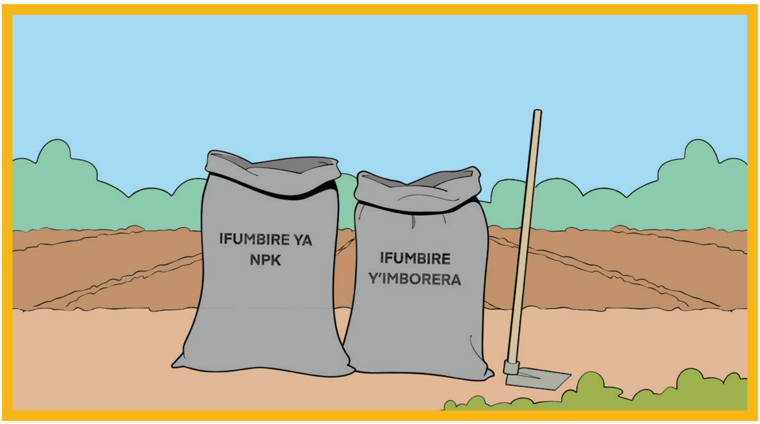
3. Crop diversification

- Plant different crops in different seasons or together (like maize and climbing beans). This helps reduce pests and diseases and improves soil fertility.

Steps on Good Soil Fertility Management Practices

Phase 1 – At Planting - Basal application

- Practice crop rotation regularly (eg, rotate beans with maize) to naturally replenish nutrients
- Intercrop cereals (maize) with legumes (beans) to improve nitrogen levels and ground cover, which improves moisture retention.
- Apply two handfuls of manure or compost directly into shallow planting holes or basins.
- Mulch soils with dry grass, crop residues (do not burn) to maintain soil moisture, prevent erosion, and improve soil organic matter
- Using the soil testing recommendations, apply the recommended basal fertilisers such as NPK and DAP.
    - It is recommended that if soil testing was not done, at the planting station, apply 2 bottle caps of fertilizer (bottle caps form a 2-litre).
    - Place manure and fertilizer in the same hole, cover with a thin soil before placing the seed



Phase 2 – Top Dressing Application: Boosting Plant growth after 6 weeks

Timing

- Apply during the first weeding, ideally 4-6 weeks after planting

Purpose

- Improve vegetative growth
- Support flowering and fruit formation
- Strengthen plant health before the reproductive phase.

Recommended fertilizers

- UREA - rich in nitrogen, promotes lead and stem growth
- NPK - Supports flowering and strong root development
- Phosphorus-rich fertilisers (DAP or TSP) promote root and flower development

Application steps

- Apply 4 bottle caps of fertiliser of inorganic fertilizer make sure that it's not very sunny.
- Place fertiliser 5-10 cm away from the plant base to avoid crop burning
- Cover lightly with soil to reduce loss from volatilisation
- Combine with weeding and light cultivation to reduce nutrient competition.

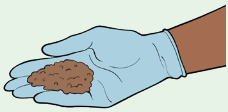
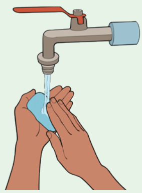
Phase 3 – Deficiency Cases

When a certain plant demonstrates signs of nutrient deficiency, fertilizer application is done on that crop alone rather than the whole field. While different crops show different signs of deficiencies, this phase will only focus on maize deficiencies. Below is a guide that illustrates how deficiencies in maize look like, and what to do in each case:

Leaf symptom	Likely deficiency	What to apply
Yellow edges 	Potassium	2 bottle caps of NPK per plant
Yellow center 	Nitrogen	2 bottle caps of urea per plant
Purple edges 	Phosphorus	2 bottle caps of NPK per plant

NB: In case of such colour symptoms observed, apply the recommended fertilisers, using the steps in phase 2.

Additional Tips for Safety and Effectiveness

Do this	Avoid this
Wear gloves when handling fertilizer 	Applying during hot sun or heavy rain
Wash your hands after handling fertilizer 	Placing fertilizer on the plant stem 
Store fertiliser in a cool, dry place (store rooms)	Using excessive fertilizer amounts
Apply when rain is expected or after rain (not during)	Applying when the soil is very dry

LAND PREPARATION AND PLANTING GUIDE FOR BEANS

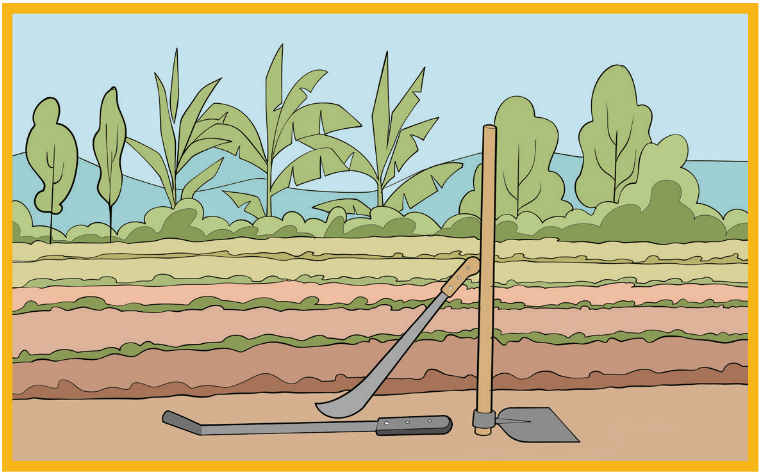
Introduction

Successful bean production begins with proper land preparation and timely planting. Beans thrive in well-drained, fertile soils with good structure and moderate organic matter. Preparing the land ensures optimal soil conditions for seed germination, root development, and plant growth. This guide provides practical steps for preparing the land and planting beans, aiming to maximize yields while maintaining soil health and minimizing environmental impact. Whether for smallholder farms or larger operations, these guidelines help lay the foundation for a productive bean-growing season.

Land Preparation

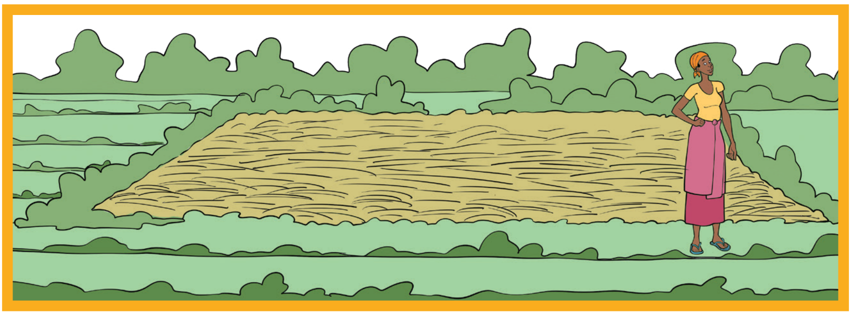
Materials Required:

Machete or slasher or a hoe.



- 2-4 weeks before planting, cut the grasses and last season's crop stalks using a machete or sickle.
    - Spread the cut material evenly across the entire field (Mulching) - (don't leave piles or burn).
    - Once dry, leave it as mulch on the soil surface to conserve moisture, suppress weeds, and protect the soil.

Tip: Do not burn or remove the plant residues — they are valuable organic matter for your soil.

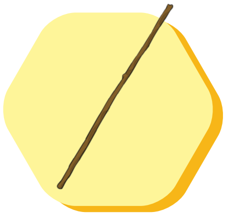


Planting Steps

Materials Required:

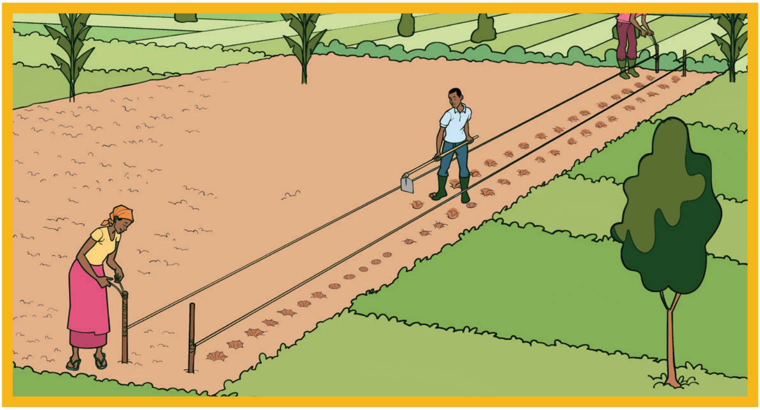
seeds, ropes, stakes, and a measuring tape.



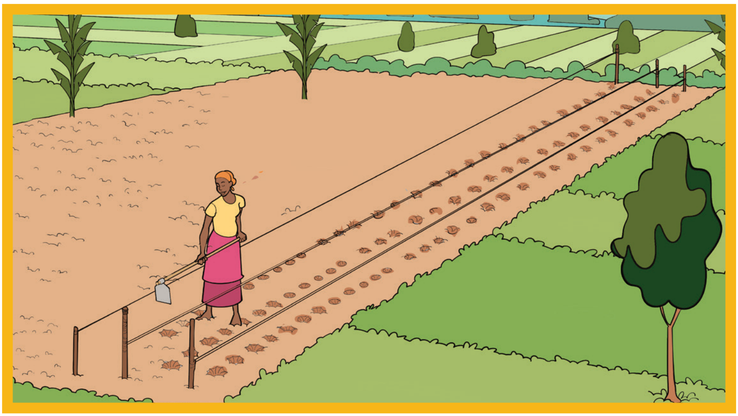
Step 1: Set Up Planting Rows - inter row spacing

- Mark the initial row using the rope and two sticks
- Using a tape measure, measure 40 cm stick, and cut it within those measurements (You will use this stick for marking the rows)
- From the first row, measure 40 cm using the marked stick to mark the second row. (Always use the rope to ensure straight lines - spaced 40 cm apart)



Step 2: Spacing between plants in rows and preparing planting holes

- Using a tape measure, measure a 20 cm stick, and cut it within those measurements (You will use this stick for marking the planting holes on the rows)
- On each row marked by a rope, mark 20 cm intervals using a measuring stick.
- At each 20 cm mark, dig a small hole about 10 cm deep and 10 cm wide.



Step 3: Add manure fertilizer and seeds

- In each hole, place two handfuls of well-decomposed manure.
- Add 2 bottle caps for basal fertilizer (see soil fertility management manual)
- Cover manure and manure lightly with soil.
- Place two bean seeds on top of the soil.
- Cover the seeds with a thin layer of soil.

Step 4: Complete the whole field by repeating step 1 to 3.

After planting

- Visit your field to evaluate germination 7-10 days from planting. In case there are seeds that did not germinate, replant as soon as possible

Remember:

Dos:	Uge wirinda ibi bikurikira:
• Leave previous season's plant residues (stalks, leaves) on the soil to serve as mulch • Only till where seeds will be planted by making holes. • Start with small plots, and upon success scale up.	• Burning crop residues. • Tilling the whole field

MAIZE LAND PREPARATION AND PLANTING - GUIDE

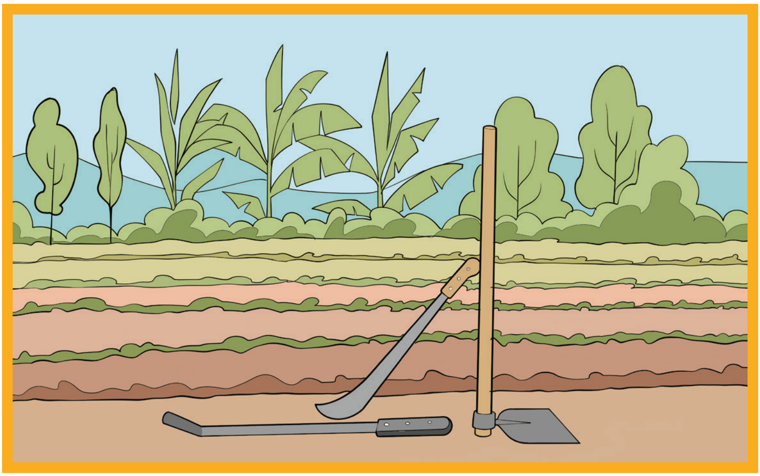
Introduction

Successful maize production begins with proper land preparation and timely planting. Maize thrive in well-drained, fertile soils with good structure and moderate organic matter. Preparing the land ensures optimal soil conditions for seed germination, root development, and plant growth. This guide provides practical steps for preparing the land and planting maize, aiming to maximize yields while maintaining soil health and minimizing environmental impact. Whether for smallholder farms or larger operations, these guidelines help lay the foundation for a productive maize-growing season.

Land clearing and weeding

Materials Required:

Machete or slasher or a hoe.



- 2-4 weeks before planting, cut the grasses and last season's crop stalks using a machete or sickle.
- Spread the cut material evenly across the entire field (Mulching) - (don't leave piles or burn).

- Once dry, leave it as mulch on the soil surface to conserve moisture, suppress weeds, and protect the soil.

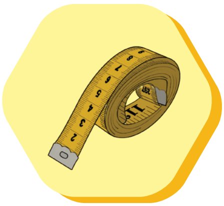
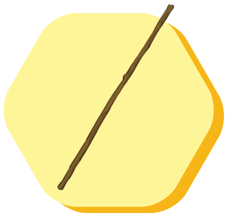
Tip: Do not burn or remove the plant residues — they are valuable organic matter for your soil.

Maize Planting Steps

Materials needed include:

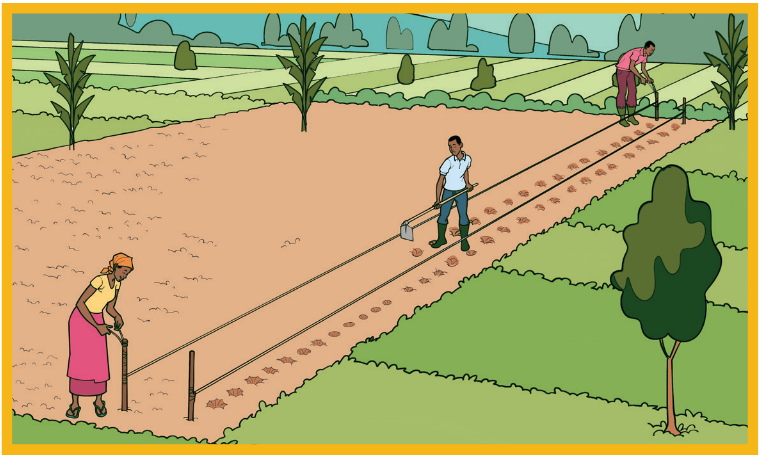
Hybrid seeds, ropes, stakes, and a measuring tape.



Step 1: Set Up Planting Rows - inter row spacing

- Mark the initial row using the rope and two sticks
- Using a tape measure, measure 75 cm stick, and cut it within those measurements (You will use this stick for marking the rows)
- From the first row, measure 75 cm using the marked stick to mark the second row. (Always use the rope to ensure straight lines - spaced 75 cm apart)



Step 2: Inrow spacing and Preparing planting holes

- Using a tape measure, measure 35 cm stick, and cut it within those measurements (You will use this stick for marking the planting holes on the rows)
- At each 20 cm mark, dig a small hole about 10 cm deep and 10 cm wide.

Step 3: Add manure, basal fertilizer and seeds

- In each hole, place two handfuls of well-decomposed manure.
- Add 2 bottle caps for basal fertilizer.
- Cover manure and fertilizer lightly with soil.
- Place two maize seeds on top of the soil.
- Cover the seeds with a thin layer of soil.

Step 4: Complete the whole field by repeating step 1 to 3.

After planting

- Visit your field to evaluate germination 7-10 days from planting. In case there are seeds that did not germinate, replant as soon as possible

PEST MANAGEMENT

Introduction

When managing pests, we focus on prevention, protection, and control. This manual will focus on general principles of pest control, with an example on maize.

Good pest management starts from seed variety selection:
Always certified seeds from reputable agro-dealers.

Techniques to Manage Pests

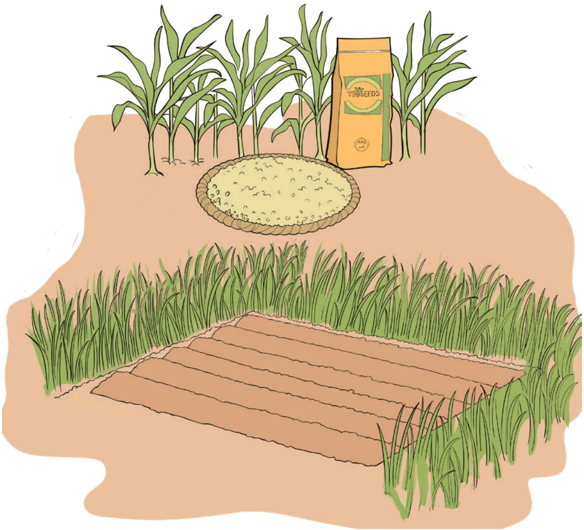
1. Gukumira

Take preventive actions to reduce pest infestation in the field.

Follow these steps:

- Crop rotation between cereals (maize) with legumes (beans).
- Use certified hybrid seeds - (**Do not use retained seed**)

- Clear and weed 1- 3 meters buffer zone around the fields
- Intercrop your maize with mucuna, and surround your whole field with napier grass



2. Protection

These actions help identify and manage the pest population early, before it becomes a major threat to productivity.

To do this, apply these measures:

- Scout your farm 2 times a week after germination, looking for signs of pest presence such as holes in leaves and larvae droppings.
- If pests are present, physically crush them.
- Install pheromone traps around the field to trap adult FAW, reducing the population.

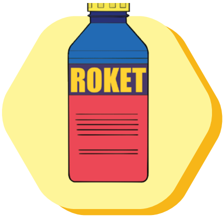
3. Control

When you observe 20% or more of the plants showing signs of pest damage, a pesticide application is justifiable. Below is a detailed guide on how to control Fall Army Worm in Maize.

Follow the following steps to apply the pesticide fo control for Fall Army Worm (FAW)

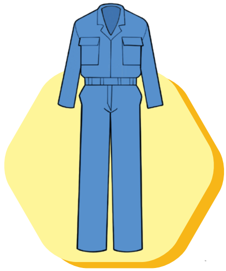
Materials Needed:

- Rocket pesticide from reputable agro-dealers.
- Knapsack sprayer (15-liter capacity)
- Personal Protective Equipment (eye goggles, respirator or a face mask in case a respirator is not available, and a worksuit (long-sleeved clothing))
- Water (15 liters)
- Measuring tools (to measure pesticides)



Rocket Pesticide

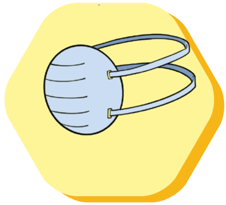
Knapsack Sprayer



Worksuit

Eye Goggles

Water (15 liters)



Face Mask

Protective boot

Safety Precautions

When working with pesticides/ chemicals, always prioritise your health and safety:

Wear Personal Protective Equipment (PPE):

- Eye goggles

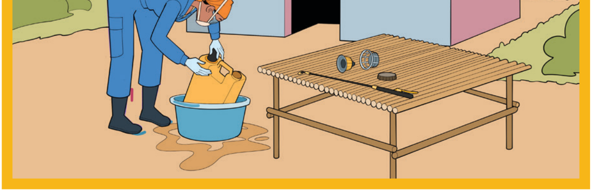
- Respirator mask (or a face mask if a respirator cannot be found)
- A worksuit or long-sleeved clothing to avoid skin contact

Steps for Applying Rocket Pesticide

1. 1Fill the knapsack sprayer with 15 liters of clean water
2. Add 45 milli-liters of rocket pesticide into the knapsack sprayer (two and a half bottle cups) - Consult a local agronomist, agrodealer or a farmer promoter if you are not sure of measurements to avoid burning your crops.
3. Spray evenly, at a low pace, paying special attention to those visibly affected by the Fall Armyworm.
4. Focus on spraying the topside of the leaves and in the whorl of the maize plant, where the FAW larvae may be hiding.
5. Ensure to spray the whole field.
6. Scout 3 days after spraying, if you observe signs of pest presence, spray again.
7. Rotate the pesticides for effective pest control.

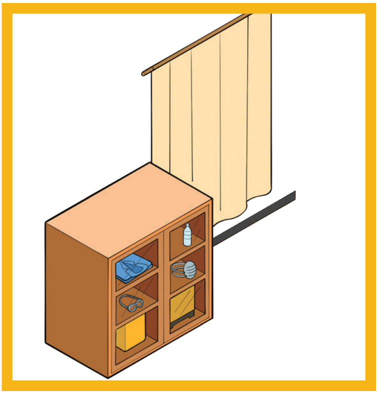
Post-spraying

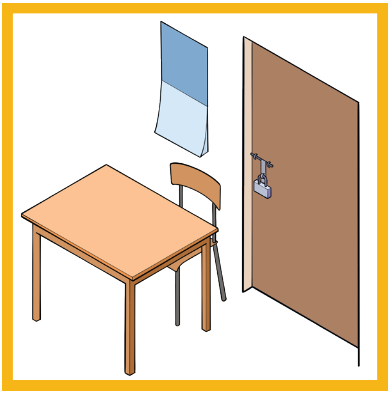
- **Once you finish spraying, rinse the sprayer with clean water at least three times after use.**



- **Make sure to dispose of rinse water inside the sprayed field to minimize environmental contamination.**

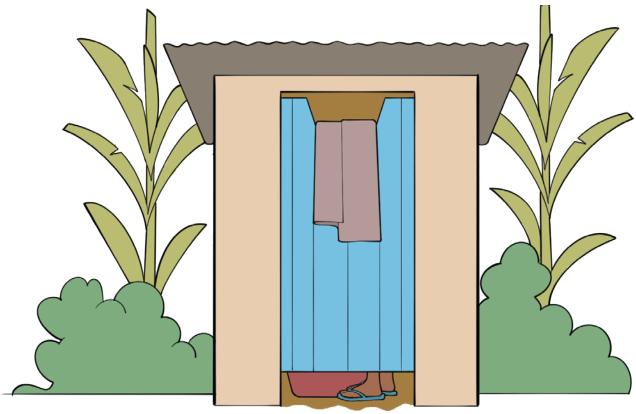
- **Store the chemicals in a cabinet with a lock/or locked store room, away from food and children.**





- **Wash your clothes that came in contact with the pesticide.**

- **Always take a shower after handling pesticide.**



Uge ukora ibikurikira:	Do not:
• Wash your clothes after spraying • Wash hands after spraying • Shower after spraying • Move at a steady pace while spraying, ensuring that you pass each plant in the field to cover it evenly.	• Wear short sleeved clothing while spraying • Spray without eye protection • Spray without a respirator on (if one can be found) • Re-enter the sprayed field within 48 hours

Remember: The earlier you act, the healthier your harvest. Prevention and protection save you money, reduce damage, and help you spray less.

Source:

- Totmak Farmers Centre, 2023;
- FAO - Manual on Integrated Fall Armyworm Managemeny - 2019

KNAPSACK SPRAY CALIBRATION TRAINING MANUAL

Nozzle calibration is a critical step before beginning any spraying activity to control pests and diseases. Correct calibration ensures that pesticides are applied evenly and at the correct dosage, increasing effectiveness and reducing waste. A clogged or malfunctioning nozzle can result in under- or over-application, leading to poor pest control, crop damage, and unnecessary cost. The Training manual provides step-by-step guidance to check, clean, calibrate, and assess your knapsack sprayer nozzle before use.

Steps to check the nozzle:

1.Test the Nozzle

- **Fill the knapsack sprayer with clean water only (without pesticides).**

- **Spray a small area in the field or on the ground to check the flow of water.**

- **Observe the spray pattern:**

  Water should come out in a steady, even mist with a consistent fan or cone shape.

- **If the spray pattern has large droplets, sporadically, or is not spraying at all, the nozzle may not be well calibrated, clogged,/or damaged.**

2.If the spray nozzle is clogged, do the following:

- **Clean the nozzle with a brush and water to remove debris or pesticide residues.**
- **Improvement test -** spray again and observe the spraying pattern to see if the nozzle can now spray in a mist form.
- **If no improvement:** remove the nozzle and check for damage or cracks.

- **If the nozzle is cracked or severely damaged:** find a replacement nozzle. You can get these at a nearest agro-dealer.

Key Reminders:

- Always calibrate before every spraying session.
- Use clean water for calibration to avoid exposure to pesticides.
- Store clean nozzles separately to avoid contamination.
- Never use sharp objects to unclog a nozzle—it may cause permanent damage.

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